Week 10: Programming and Research

Programming task

Optional targeted Python review.
Required: operator demo and independent project verification.

Research task

No new research assignment.
Complete the existing final handoff.
No additional coding project this week.

Open learning_materials/Week_10.ipynb. Follow Start Here; keep your work in that notebook. Email: submit.progmis@gmail.com.

Programming task

Optional review only; protect both final project sessions.

1. Optional: run versus import

Do this

Choose only if startup is unclear.
Use the Session 1 guarded main example.
Predict direct execution and import.

Observe and explain

In a separate copy, remove the guard.
Observe unwanted startup.
Restore it and rerun.

2. Optional: parse an argument

Do this

Open #session-2: Predict → Run.
Try argument text 3 and 0.
Then try many in a copy.

Observe and explain

Both 3 and 0 parse as int.
many causes a parser error.
Explain conversion versus validation.

3. Optional: reset versus retry

Do this

Use the Session 1 retry example.
Predict a second action without reset.
Compare with a fresh starting state.

Observe and explain

Fix only a problem you encounter.
Record the difference and explanation.
Stop when your project run route works.

Programming practice: done checklist

Keep in your notebook

Prediction before running
Your changed or repaired code
Observed output or error
One explanation of why it happened

Next step

Optional exploration can wait.
Save the notebook and a backup.
Continue to the project application.
Submit to submit.progmis@gmail.com.

Optional review only. No new programming topic or extra project.

Required Session 2: operator demonstration

Activity	What to do / check
Prepare	Assign stations and roles; give the peer exact reset/run instructions.
Demonstrate	Parallel demos with role reversal: baseline and approved feature.
Peer check	Peer tries an unseen input and asks a coordinator question.
Save evidence	Save first difficulty, guide revision and retest; retain outputs and failure/event checks.

Open #operator-demo and #project-task. Optional Python review must not displace these tasks.

Required Session 3: individual verification

Activity	What to do / check
Independent task	Receive the fresh variant; predict or change the assigned rule independently.
Written reasoning	Explain expected/observed results and your decision in writing.
Individual explanation	Oral spot-checks or instructor sampling; then save and submit privately.
Final handoff	Approved feature, reset/run/recovery guide, outputs, peer revision/retest and individual explanation.

Open #project-studio. Rehearsal P06 is unreviewed; the final variant may differ. No extra feature required.